AI BASED CANDIDATES RECOMMENDING SYSTEM FOR A COMPANY VACANCY

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Project Proposal Report

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DECLARATION OF CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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ABSTRACT

Evaluating applicants' interpersonal skills in an interview continues to shape critical decisions in recruitment in the current job market. Our research, "Soft Skill Analysis for Candidates During the Interview," develops a system for assessing behavioral tendencies and the confidence to which a candidate possesses and draws upon using hand gesture recognition. This research differs from the traditional engagement-based approaches in that it applies tracking and deep learning based gesture recognition to focus on the detection and classification of candidates' hand movements and gestures. The system makes a distinction between positive gestures (e.g., open hand movements, upright posture) and negative gestures (e.g., fidgeting, defensive arm crossing) that indicate a person's non-verbal communication and confidence. By analyzing the number and the types of gestures which are associated with some soft skills, we compute confidence and behavioural congruence of candidates during the interviews. Preliminary results indicate that the analysis of hand gestures greatly enhances the interpretation of non-verbal communication and, when combined with the traditional task-based performance metrics, enriches the overall assessment of a candidate. This research is aimed at the automation of gesture analysis for recruitment purposes using AI technologies which is the first step towards more objective and smarter recruitment.

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1. INTRODUCTION

Today’s job market not only needs to take into account candidates’ technical skills, but much more often than in the past, employers place equal, if not more importance on soft skills for long-term workplace success. Characteristics like confidence, flexibility, and morale, along with soft skills like communication, leadership, and interpersonal awareness, are important factors in predicting how well an individual is likely to succeed in teamwork environments, leadership roles, or customer-facing roles. Often, these soft skills can be a deciding factor between candidates with similar technical skills.

Soft skills are difficult to evaluate consistently in two-dimensional traditional recruitment contexts. Conventional interviews tend to be based on subjective assessments that draw heavily on an interviewer’s perceptions, biases, and non-verbal behaviours. With this approach, there are inconsistencies and unfairness in decision making that can affect candidates, who "could have been misjudged, not because of lack of evidence, but because of unconscious bias" (Robles, 2012). Therefore, more organizations are leveraging technology-enabled assessment frameworks that are less prone to subjective interpretation, which will add fairness, accuracy, and expected data insights for organizations hiring people.

NVC is one extramoral significant area of interest would be NVC analysis (Suggestion of the analysis of non-verbal communication, especially hand movement, and gestures). With verbal answers, even when conducted as a live interview there are chances for rehearsed responses, gestures however occur unconsciously and openly in an interview. Therefore, while analysing candid responses, look at the gestures because the truth will most commonly lie in the gestures. For example, palms open, movements steady, spontaneous and restricted to a relaxed environment indicate gestures indicative of self-efficacy and openness of a person. Alternatively, fidgeting hands, clenched fist(s), and crossing limbs might suggest nervousness, insecurity or lack self-efficacy and a different disposition. More structured methods of NVC analysis help note these actions to develop an estimate of a potential candidate's probable confidence during an interview, this is important in softness skill assessment.

Today, with the increasing availability of computer vision technologies and machine learning models, this kind of behavioral analysis is practical and scalable. Using existing frameworks such as OpenCV for hand tracking and advanced classifiers for gesture recognition, systems can capture slight motion patterns, extract relevant features, and expect confidence levels in real-time. As the ability to seamlessly integrate gesture-based behavioral analytics into the recruitment pipeline develops, organizations can expect to take a more data-informed approach towards assessments, in addition to traditional evaluations. If possible, the measurements outcomes should help organizations be fairer, while also better understanding candidates’ non-verbal communication capabilities, and postures within behavioral tendencies.

1.1 Background Literature

The importance of non-verbal communication has been recognized for many years in psychology, communication, and organizational behavior. Researchers agree that gestures, facial expressions, postures, and other body language are important indicators of a person's emotional state, confidence, and interpersonal orientation. Birdwhistell (1970) noted that up to two-thirds of communication could be considered non-verbal, while Mehrabian (1972) emphasized that tone of voice and body language might matter more than the words themselves in forming impressions. Although the exact proportions are still debated, these early works established the importance of gestures as more than incidental behavior and the centrality of gestures in interpersonal communication.

Expanding on this foundation, McNeill (1992) proposed that gestures are part of speech and thought, and thus they reveal information about speaker's level of confidence and certainty. Kendon (2004) expanded on the social dimension of gesture, showing that expansive gestures suggest signals of confidence, while closed or defensive gestures can signal discomfort or insecurity. Numerous empirical studies have categorized gestures in more concrete ways as well: confidence-enhancing illustrators include smooth outward movements and open palm gestures that often underscore the speech; while confidence-reducing adaptors, such as fidgeting, hand-to-face gestures, and tightly locked hands — typically with other behaviours signifying anxiety (Ekman & Friesen, 1969). In organizational contexts, Burgoon et al. (1996) documented that non-verbal cues powerfully influence how credibility and leadership is perceived, and relationally they are pertinent to hiring and professional performance assessments.

Interest in workplace effectiveness has grown, and research has emphasized confidence emergence as a measurable soft skill. Confidence, both personally and professionally, has an impact on audience perception of others as well as workplace performance with leaders and team members and in customer-driven context. Research has shown that interviewers rely on candidates' body language, automatic responses, and normally unconscious hand and arm action to assess confidence during the interviewing process (Argyle, 1988), and this assessment has been largely subjective. Both acknowledgments of this knowledge raise the question of how new, objective, technology-centered, practiced evaluation measures could offer a more objective means of assessment and how the inclusion of soft skill based confidence could benefit decision makers.

The growth of computer vision and artificial intelligence (AI) has opened up the possibility of automating this type of behavioral analysis. The earliest computational non-verbal communication approaches focused primarily on face recognition and affective computing. For example, Kapoor and Picard (2005) created frameworks for real-time detection of emotional engagement based on facial and postural cues while Zeng et al. (2009) summarized the techniques for automated facial expression recognition revealing the ways visual features could be assigned to emotional states. These pioneering works had a significant implication for the context of affective computing, however they paid little attention to hand gestures, which can serve as an equally or sometimes more revealing indicator during social interactions.

At the same time, the area of gesture recognition started to develop as a field, principally enabled by new developments in computer vision. For example, Bradski (2000) showed OpenCV, a pivotal and prominent resource in the area of gesture/motion analysis. Then, Molchanov et al. (2015) used convolutional neural networks (CNN) and recurrent architectures for dynamic hand gesture recognition and successfully classified these gestures despite variations in environments. The technological pipeline to this analysis is complicated: using computer vision libraries (i.e., MediaPipe) to first extract 21 keypoints of the hand, the libraries would transform the keypoints into numerical data streams. Then, these temporal sequences of coordinates could be analysed by machine learning models (temporal convolutional networks (TCN)), to extract features like velocity, trajectory, and spatial extent of these movements and classify them into discernible patterns. Rautaray and Agrawal (2015) again provided a comprehensive review of these systems while showing the potential for hand movements to give a degree of reliability to signify cognitive and emotional states. These developments provide a solid technological background and leave a clear pierced pathway for the application of gesture recognition in real-world applications such as recruitment.

Many domains have already provided a case for the added value of gesture analysis. In education, gestures have been correlated to attention and engagement. Murthy and Jadon (2009), for example, found that the frequency of hand raising combined with active (restless) body movements can forecast student participation in the classroom. In healthcare, gestures are monitored to contribute to neurological assessments, where hand movements defined as repetitive or uncontrolled are linked to stress, anxiety or cognitive decline (Sethi & Ranganath, 2017). In human-computer interaction (HCI), gestures are a natural input for control of systems, which indicates the reliability of gestures as behavioral signals (Chen et al., 2020). These different areas of study further evidences the validity of gesture recognition to reliably observe behavioral patterns offering significant potential for its use in recruitment.

Despite this advancement, research in recruitment has only recently investigated AI-based non-verbal marks. Most of the available studies tend to be either facial expressions based recognition (or for voice - sentiment analysis) (Pentland, 2008; Pantic & Rothkrantz, 2003). Although they offer important messages, they do not address some of the quieter, but very informative, hand gestures. The hands, in particular, can convey important information about the candidate's confidence level and a solid understanding of the underlying process of creating hand gestures may be, in fact, important valuable signals in understanding confidence. Fluid, open movements may suggest ease and self-assurance while fidgeting movements and closed postures could indicate hesitance and/or unnecessary umbilical fidgeting. Importantly, while facial expressions can be sometimes planned, conscious or directed forms of communication, gestures remain a great way to assess the unconscious dimensions of a social communication. Hand gestures often are an unconscious channel, which is why gestures on the part of an interactor may be understood more directly and as being more reliable to assess confidence.

Moreover, there has been an increasing use of AI-assisted hiring applications, indicating the immediacy of the hand movement analysis. In the field of organizational psychology, research has indicated (Huang & Ryan, 2011) that the non-verbal behaviors of candidates have a direct impact on the ratings interviewers give, particularly ratings on perceived confidence and leadership potential. However, traditional assessment methods are highly reliant on perceptual factors, and thus, are often subject to perceptual bias. Using gesture recognition technology as a assessment method would offer an objective and quantitative means to assessment perceived confidence, thereby increasing the fairness and consistency of hiring decisions. However, it is important to point out that we must acknowledge the ethical implications of our quest for objectivity. Algorithmic bias is the primary risk; models that are trained on data that misrepresents the characteristics of the population may eventually misinterpret the gestures of certain genders, cultures, or even ages, or that people with limited physical abilities alone could introduce the bias, even exacerbating discrimination. Thus far, we should be developing these tools based on algorithmic fairness principles regarding the diversity of training data and audit frequency.

Recent advances in deep learning allow us to go beyond a binary accuracy of gestures, towards more contextual interpretations. For example, in Liu et al. (2020), authors described multimodal frameworks which combine hand gestures (manual) with speech (vocal) and facial (visual) contextual cues in order to reach a decision for emotion recognition. This multimodal framework demonstrated that gestures can add independent predictive power with gesture-specific classified outcomes for context such as confidence associated categories. We can apply these or other multimodal frameworks to recruitment, so we can create systems that would classify gestures into positive (confidence-enhancers) and negative (confidence-depleters), and in turn, use these to predict overall confidence levels in an interview. In the future of this area, these multimodal systems could examine holistic systems to consider all forms of gesture data including (vocal prosody from the spoken word and embodied posture); not to replace human judgment, but as an adaptive, robust, aware system to help mediate some of the inherent bias in people's perceptions and decisions to add depth and equity for the future process of recruitment.

1.2 Research Gap

Interest in the study of soft skills in relation to recruitment has expanded, particularly as organizations are more aware that cognitive ability measures alone cannot predict future long-term success. Non-verbal behaviours, such as facial expressions, body posture, and hand gestures, are remarkably powerful indicators of candidates' emotional state, confidence, and interpersonal skills. However, the use of computer vision and AI for analysing elements of non-verbal communication in recruitment contexts is still an emerging field of study. Most existing work has focused on emotion recognition or facial engagement tracking, meaning that analyses of hand gesture as a form of non-verbal communication has remained relatively undeveloped.

1. Underutilization of Hand Gesture Analysis in Recruitment

Hand gestures are an extremely important aspect of non-verbal communication, frequently occurring unconsciously, and thus revealing an authentic glimpse into individuals' psychological states. As McNeill (1992) explains, hand gestures are not simply verbal fillers, but they also represent the cognitive and affective states of individuals. For example, an open palm might represent confidence and honesty, while establishing enough eye contact with the interviewer, fidgeting, and repeating the same motions are possible signs of anxiety and lack of preparation. Nevertheless, even with this evidence, research that emphasizes verbal fluency and/or facial expressions of emotions, such as from the recruitment perspective, ignores hand gestures as a quantifiable and direct indicator of confidence.

The technological framework for analyzing such gestures is now possible but not widely applied. The technology available today would use pose estimation models (e.g., MediaPipe or OpenPose) with the goal of extracting precise skeletal data (21-points) for each hand from video interviews. Time-series features such as the velocity, acceleration, smoothness (jerk), and spatial distance of movements would be derived from the coordinate data, and the most relevant machine learning classifiers would be temporal convolutional networks (TCNs) or Long Short-Term Memory (LSTM) networks to classify temporal sequences into recognizably interpreted patterns, such as "illustrative gesture" and "nervous fidget." While there is access to relevant technology, there are few efforts to systematically capture, quantify, and derive meaning from these features that create a standardized confidence assessment to be used for recruitment, thus presenting a gap between the technological and HR practicalities.

2. Lack of Formal Gesture-to-Confidence Models

Research in gesture recognition is predominantly effective in areas such as sign language interpretation (Rautaray & Agrawal, 2015), human-computer interaction (Molchanov et al., 2015) and learning analytics (Murthy & Jadon, 2009). Researchers use computer vision and deep learning methods for gesture detection and classification, with a reasonable level of accuracy. However, the mapping of low-level gesture characteristics to higher-order psychological constructs, like confidence, is less developed.

For models to work not only to classify but to score on a continuum of confidence we require a training dataset that is composed with labeled specific gesture features (e.g., high speed + high smoothness + high distal extent) and correlated human judgement of overall confidence using Expert-Annotated Ground Truth or Crowd-Sourced Perception Evaluation. Otherwise, any interpretation of gestures or confidence behaviour on the continuum between 0 and 10 is anecdotal. Indeed, aspects of gesture and confidence are discussed in the psychology literature (Kendon, 2004; Argyle, 1988), yet a computational model has not been developed and validated to document the feature to behaviour mapping, in a recruitment context.

This is where there seems to be a gap; delaying the advancement of gesture recognition as an innovative technology that meaningfully intersects with the assessment of soft skills.

3. Cultural and Contextual Variability in Gesture Interpretation

Gestures are often affected by cultural background, situational contexts, and each person's own individual behaviors. For instance, Western contexts often associate steady hand gestures that utilize open palms with confidence, while others may see restricted, deliberate gestures as a respectful representation of discipline (e.g., some East Asian contexts). Similarly, certain habits that may indicate nervousness, such as tapping a table, may not necessarily represent lack of confidence but rather more accurately represent a coping mechanism or cognitive trigger for that individual.

Most existing gesture recognition systems are trained on regional datasets that do not normalize for cultural adaptability or interpretation in the context of the interview. This knowledge gap presents a concern for normalizing gesture based confidence analysis frameworks, which are attempting to be generalized to a diverse, global candidate pool. The only way to be truly objective is for a system to either be calibrated based on a multinational dataset or include a module for contextual calibration whereby the baseline gesture behavior is established for each candidate in low-stress introductory parts of each interview. Otherwise, gesture analysis does possibly categorize cultural biases that are systematic in representing individuals as a deliberate approach for objective recruitment analytics while raising serious ethical issues.

4. Absence of Weighted, Multi-Dimensional Scoring Systems

The role of hand gesture analysis in determining candidate confidence has been emphasized in the proposed framework. However, hand gestures represent only one part of the larger communication process. Person-to-person communication consists of a number of modalities including, but not limited to, hand gestures. Confidence is not defined by hand gestures alone, but is an amalgamation of voice, eye behavior, and hand gesture behavior.

To address the variability of confidence, the proposed framework provides a flexible scoring regime that allows the degree of instrumentality of the hand gesture scores to be set as HR managers deem appropriate based on the position they are recruiting for. The unlimited configurability is critical to predictive validity, specifically whether the proposed framework accurately predicts on the job success in the case of a specific role.

For example:

• In customer-facing roles (e.g., sales, negotiation, leadership), where expressive non-verbal communication is critical, HR may assign a higher weight (e.g., 30–40%) to gesture analysis.
• In technical or analytical roles (e.g., software engineering, data science), where communication style may be more subdued and rely more on verbal clarity, a lower weighting (e.g., 15–20%) may be more appropriate.
• For roles requiring balanced soft-skill evaluation (e.g., project management), an intermediate weighting (25–30%) could be applied.

Thus, hand gestures are not treated as an absolute determinant but as a configurable parameter within the overall confidence assessment.

Composite Confidence Scoring

The overall confidence score (OCS) is derived as a weighted sum of the three components:
OCS = (W_v × C_v) + (W_e × C_e) + (W_h × C_h)
Where:
• **C_v** = Confidence score from vocal features (e.g., tone, pitch, speech rate, clarity derived from audio analysis).
• **C_e** = Confidence score from eye behavior (e.g., gaze stability, blink rate, proportional eye contact measured via facial landmarks).
• **C_h** = Confidence score from hand gesture analysis (e.g., based on spatio-temporal features like gesture entropy, smoothness, and type classification).
• **W_v, W_e, W_h** = Configurable weightings decided by HR based on role requirements, such that: **W_v + W_e + W_h = 1 (100%)**.

Implementation of the Scoring System:

The system would function through a **dashboard interface** for HR professionals. After the AI processes an interview video, it generates the three component scores (C_v, C_e, C_h). The HR manager can then either:

1. **Use a Pre-set Template:** Select a role profile (e.g., "Sales Lead," "Back-End Developer") which automatically loads the pre-defined weightings for that role.

2. **Customize Manually:** Manually adjust the sliders for each weight based on the specific needs of the role and organization, allowing for maximum flexibility.

Component	Weight (Example 1: Technical Role)	Weight (Example 2: Balanced Role)	Weight (Example 3: Leadership Role)
Voice (C_v)	35%	35%	30%
Eye Behavior (C_e)	45%	35%	30%
Hand Gestures (C_h)	20%	30%	30%
Total	**100%**	**100%**	**100%**

The flexibility, multimodal nature, and ethical considerations of this approach allow for the contextualized contribution of each non-verbal channel to be included, making the final confidence score a better, fairer, and more valid indicator of quality for hiring decisions. Future work must validate this model through longitudinal studies correlating OCS with actual job performance metrics across job roles.

Technical and Practical Challenges

- **Occlusion and camera quality**: Candidates may inadvertently occlude their hands by resting them on the desk, crossing their arms, or gesturing beyond the viewing area of the camera. Many interviews are completed with consumer-grade webcams that have low resolution or frame rates, which further diminishes the ability to detect covert hand or finger movements. Even when using high-definition or upgraded webcams through video conferencing, the compression from the streaming technology within the platforms could hurt the gesture data even more.

- **Real-time tracking and computational cost**: High-accuracy gesture recognition models, particularly those leveraging deep learning and fine-tuned neural networks, are typically computationally expensive. Therefore, even if the model is accurate, if the model can only run at 30 frames per second with an indicator with a 200ms lag, the model will fall behind once the latency exceeds 200ms. This could result in realizable delays to candidates in standard laptops, as well as cloud-based systems in any particular recruitment tool, at which there may be multiple interviews taking place at once. Real-time accuracy requires a forward-thinking combination not only on the device but also the computational infrastructure and costs associated with capturing gesture data from lots of people simultaneously.

- **Environmental variation**: Gesture recognition applications are fragile to the candidate across different candidate environments, since there could be variable levels of ambient light (ex. backlighting by a window, poorly lit room at the time of the interview), as well as potential occlusions that can be produced from suboptimal camera angles (eg. laptop positioned too low, distorted picture due to wide-angle view), or simply having a multi-faceted contest thus maybe the case of distracting backgrounds or clutter, all contribute to noise in the data. This is no exaggeration: as we discussed, there is no standardized gesture analysis that will work across entirely different candidates, especially when multiple candidates are involved in remote interviews with an uncertain environment.

- **Behavioral noise and ambiguity:** Not all gestures will represent psychological states such as confidence and nervousness. Many hand movements are commonly exhibited behaviors as opposed to purposeful, communicative gesturing, including scratching; fidgeting objects; glasses adjustments; or resting chin. It is a non-trivial classification task to determine the exemplifying cues from the behavioral array of irrelevant movements. The cultural variation of gesture use further complicates the matter, particularly since the same gesture may mean something completely different in different countries and regions.

- **Limited datasets:** Building a reliable gesture-recognition system requires large and diverse datasets collected under realistic interview settings. Although, most existing datasets that have similar gesture-recognition goals are small datasets collected in a laboratory setting that have little ecological validity. This restriction limits a model's ability to generalize to real-world recruitment platforms.

- **Ethical and fairness concerns:** Even if gesture recognition was technically reliable, utilising this technology in recruitment presents bias, privacy, and candidate acceptance issues. While over-applying the importance of physical expressiveness, neurodivergent individuals or culturally-based hiring criticalities regarding gesturing may present a disadvantage when making decisions on the fairness of assessments.

Because of these technical, contextual, and ethical barriers, few studies attempt to address gesture analysis in practical, high-stakes interview scenarios. As a result, much of the research remains theoretical, with limited translation into scalable, trustworthy recruitment platforms.

1.3 Research Problem

Measuring confidence in job interviews has been an unstructured, subjective process for years. Experienced recruiters, for example, make use of observable cues, such as tone of voice, eye gaze, and hand gestures to assess confidence, and determiners of other behavioral indicants, but assessors are subject to their own personal views and cultural interpretations when assessing and incorporating observable cues, and this leads to inconsistencies in hiring decision-making. Research findings in the fields of psychology, as well as research on organizations, establish that that nonverbal communication carries significant amounts of the meaning in interpersonal interactions (Mehrabian, 1972). Specifically, while many cues represent aspects of communication, hand gestures offer the most powerful, and underutilized estimate of confidence and poise.

The core research problem is that while gestures are clearly significant in estimation of perceived confidence, there simply does not exist a standardized, validated computational modeling framework that can automatically detect, classify, and quantify gesture-based confidence in real-world job interview contexts. The related, but broader research problem is to accurately calculate fair, interpretable, and composite confidence score using gesture and other modalities (voice, gaze) as features, and to build a model that may be transferable, so that fundamentally, the problem is both how we design and build a computational modelling framework that resolves content capture across jobs and cultures.

So, the research problem can be divided into 5 critical dimensions:

1. Robust Detection and Classification of Hand Gestures in Real-World Environments

At its core, the technical challenge is to build a model agnostic to:

- **Different Camera Angling and Quality**. Gestures will need to be detected from the standard low quality webcam video feed, which may include low resolution content and highly compressed, and may be even further confounding by the angle at which the camera is looking at the candidate (e.g., a laptop camera directed up at the candidate).

- **Occlusions and Lighting**. There are occluding factors (e.g., a desk) can partially block the view of hands on the candidate. Alternatively, lighting conditions may obscure keypoints based on dynamic lighting direction, dynamic shadowing or backlighting.

- **Micro Gestures and Ambiguity.** The system will need to resolve both macro level illustrators; the large gesturing associated with speech and micro level adaptors; small fidgeting movements such as self-touch. All of which confound candidate, non-verbal confidence planning. For example, an action such as adjusting glasses must be distinguished within context from the behavior of putting a fidgeting hand on one's face as a sign of stress.

This creates the research problem of designing a solid pipeline that employs the best state-of-the-art pose estimation architectures (e.g., HRNet, designed for high-resolution keypoint detection) and spatio-temporal deep learning models (e.g., Graph Convolutional Networks can model the skeletal relationships of the hand over time) that have been trained and fine-tuned on a rich and included interpretation video dataset to accommodate these natural fluctuations.

2. Quantitively Mapping Gesture Dynamics to Confidence Constructs

The issue is bigger than just detection - it lies in how to appropriately interpret the detection and give it meaningfulness. It will be critical to go from label (e.g. fidget) to meaningfully identifying a viable feature representation that quantitatively estimate a confidence metric. To quantify, meaningful constructs/estimates of psychological constructs we need to derive and compute a set of features that can be derived from the raw point data that have on relationship to psychological states. Our feature sets are:

- **Kinematic Features:** The velocity, acceleration, jerk (and smoothness - jerk) provide features around hand motion. Confident movement for the hand will be smoother and maybe deliberate. Nervous movement for a hand will be jerkier and erratic.

- **Spatiotemporal Features:** The spatial extent of the gesture (the total 3D volumetric size the gesture takes up), and the trajectory of the gesture defined by time. More confident people will use larger, expansive use of their hands.

- **Entropy, Predictability:** The unpredictability of the hand movement. There was high entropy and unpredictability with my hand movement. While sometimes hand movement was low random and, beat-like rhythm which expressed calm or focus.

The research problem is developing a validated, ground-truthed mapping function $f(G) \rightarrow C\_h$ where a vector of gesture features (G) is mapped to a normalized confidence score (C_h). This involved developing a labeled dataset through the participation of an expert psychologist and human resources professionals annotating video clips for perceived confidence. From this, a machine learning model (for example a Support Vector Regressor or Neural Network), can learn the complex, non-linear relationships associated with low-level features and high-level perception.

3. Developing a Context-Aware, Multimodal Fusion Framework

Confidence is a multi-faceted construct, and using a single modality to assess confidence is reductive. The real research challenge is to how to fuse voice (paralanguage) information, eye behaviour and gestures in order to produce a unified, valid and reliable confidence score. This is not as simple as a weighted average but rather a complicated fusion problem because of:

- **Temporal Asynchrony:** A confident gestural time stamp may happen a second later than the confident vocal emphasis. The unified framework must put these modalities in coordination temporally, possibly using models like Multimodal Transformers which are able to learn cross attention between audio and visual information sequences.

- **Modality Reliability:** The system must weigh the cadence of modalities differently based on the modality's unique reliability at any point in time. If the camera view is occluded for example, fusing of gestures should weight automatically to a reduction and therefore, the vocal modality should be weighted appropriately to increase.

- **Incongruous Information**: The model must be trained to resolve competing signals (a steady voice but nervous fidgeting) in a human-like way unwinding the conflicts likely by weighting more reliable or more relevant information to the context, In this case likely justification for discomfort would outweigh any attributes of disguise.

The research question becomes how to design an adaptive fusion architecture that accounts for all of these components as opposed to a static formula, and can progress from a uninformed method of a composite confidence score (OCS) to a dynamic adaptive replication of the human experience.

4. Interface Design for Flexible and Explainable HR interface with Fairness Constraints

The usefulness of this technology is in whether HR can use it. The research problem is really two problems:

- •Configurable Weighting Interface: The first is a simple dashboard design where HR managers will be able to input the role-specific weights (Wv, We, Wh) initially and be able to view the evidence provided. For instance, if I click on a final score, I would want to see detail such as "The high vocal score is due to low filler words; the moderate gesture score is negatively impacted by a pattern of tapping / puckering on the desk".

- Bias Design: The second problem is more difficult and comes to algorithmic fairness by design. The design must include:

- Cultural Calibration: using enter questions prior to the interview, or a brief calibration video, to develop the baseline gestural/behavioural style of the candidate.

- Accessibility Overrides: HR must have the ability to switch off the gesture module entirely for candidates who indicate they have a physical disability or be able to re-normalize based on a declared cultural background.

- Bias Auditing: continually testing the model for subgroup fairness - this would mean ensuring that average confidences scores did not systematically differ across gender or ethnicity groups as a result of an artifact of the training data.

This converts a technical problem into an HCI and ethical AI problem, which prevents us from just replacing human judgement with a more accurate scoring tool.

5. Ensuring Real-Time Performance and Scalable Deployment

To be of practical use, the system needs to provide feedback in nearly real-time performance in a video interview. This poses a significant engineering research problem:

- Computational Efficiency: The model needs to be optimized, so it can be run on typical hardware. This problem may require using model distillation so a smaller and faster student network can be derived from the larger teacher model, or the architecture can be pruned and quantized to reduce the computational performance with minimal accuracy recourse.

- Architectural Decision: Establishing how to run the architecture as cloud-based API (which uses much more power but has the luxury of not requiring the transfer of data) and it can either be done on-device (in terms of privacy) but a very efficient model. Its also possible that there is a hybrid where only anonymized confidence metrics are uploaded after an on-device invocation of the core model.

Summary of Core Research Problems

Taking all of these aspects together, we can we summarize the ultimate research problem as:

How can we create a strong, fair, and interpretable AI system that uses advanced computer vision and multimodal machine learning to create an objective, quantifiable confidence score from subjective hand gestures in conjunction with vocal/ocular cues to produce a tool, scalable in real-time, for HR professionals to help improve interview assessment?

By breaking this down into these technical, ethical, and practical components, this research will not only lead to a robust AI-driven confidence assessment system but lead to a system that is practical value and is ethically responsible to modern hiring procedures.

1.4 OBJECTIVES

1.4.1 Main Objective

This research intends to create and install an artificial intelligence framework that reviews hand gestures during interviews to objectively assess the candidate's confidence level and to associate this information with other behavioral signals, such as the voice and the eyes. This allows for interpreting gestures, which are typically considered subjective qualitative observations, into objective, quantitative measurable and verifiable confidence indicators for recruitment decision making. Hand gestures will represent between 20%–40% of the level of confidence score (depending on the HR role and preference of application). By allowing hand gestures, as a vital but often overlooked behavioral sign, to be incorporated as an important part of soft-skills assessment.

1.4.2 Specific Objectives

1. Detection and Tracking of Hand Gestures

The first goal is to be able to detect hand movement and track it accurately whether during live or recorded interviews. Hand gestures are extremely dynamic and vary considerably from person to person regarding speed, shape, and location, making it difficult to consistently capture. For the detection objective, we will utilize computer vision techniques such as OpenCV to track motion and MediaPipe to detect the key-points of the hand joints and the position of the fingers accurately.

Moreover, it will be imperative to distinguish between hand movements that are intentional and expressive versus unintentional hand movements or movements that may be derived from nervousness, as important confidence inference depends on this distinction. For example, gestures may be a stable sequence of hand motion aligned with the speech, while no gesture or erratic gestures may need to be made about nervousness. The model will track parameters, including, for example, rate of movement in terms of the frequency of gestures, the trajectory of gestures, and focussing on whether the gesture was stable throughout the attributed time of the interview. A unique contribution will be making a robust tracking pipeline to work in real-time, across various interview conditions, specifically: variations of lighting, camera, and background.

2. Classification of Positive vs. Negative Gestures

The second goal is to develop a classification model that helps differentiate positive and negative gestures as well as helpfulness-like labels related to confidence. In addition to the behavioral psychology and non-verbal communication studies used to assess speech, I will capitalize on three classes of gestures, including:

- Positive gestures: open palm, fluid gestures, managed hands, matching on gestures with speech
- Negative gestures: excessive fidgeting, wringing hands, clenched fist, repeated touching themselves, or defensive bodily crossed up
- Neutral gestures: limited or resting movements that are not particularly confident indicators.

To classify subtle differences in time-based gestures I will utilize machine learning methods such as Random Forest classifiers and deep learning sequence based models, (e.g., CNN-LSTM) because confidence is present in gestures over time, not in one-off movements. For example, a candidate can have a fleeting neutral moment adjusting their hands, but minutes of consistent fidgeting would be classified as negative and reduce confidence scores.

The classification stage is critical because it converts raw movement data into **interpretable behavioral categories**, bridging the gap between visual tracking and psychological meaning.

3. Scoring of Hand Gesture Contribution to Confidence

After gestures are categorized, the next step is to create a quantitative scoring scale where hand gestures are given a defined percentage (e.g., 20-40%) of the overall confidence. This is truly a new innovation because it transitioned from qualitative HR introspection to numeric scoring. Each gesture (positive, negative, neutral) will have a confidence weight assigned. Candidates' will be measured on a confidence weight scored on a set scale (e.g. 0-100) related to the total number of gestures, their duration, and the quality of each gesture.

More importantly, a gesture contribution percentage weight can be configured by HR. For example, in a physical customer service job, HR may choose to have a percent weight of 40% for hand gestures. For an IT technician role, the gesture contribution percent weight may only be 20% because clarity of speech is more relevant - a gesture may just not matter. Regardless, the model gives HR the ability to create applicable metrics for their industry or context of the interview.

As a result, the score model converts gesture contributions into a standardized metric for evaluation as well as create a consistent measure of influence that gestures have on confidence metrics for a total participant assessment that can be used in the selection process.

4. Integration into a Multimodal Confidence Framework

The fourth goal is to associate gesture-based scores with other behavioral modalities--voice metrics and eye metrics—to form an integrated multimodal confidence framework. Research suggests that confidence is rarely expressed in one channel, but rather manifests with the congruence of verbal tone, eye contact, and gestures. This goal ensures that gesture based data is not analyzed in a vacuum, but rather relative to these companion signals.

The integrated framework will consist of:

- Voice features: tone, pitch stability, speech fluency, and filler words.

- Eye features: gaze direction, blink rate, eye contact duration.

- Hand gestures: frequency, steadiness, open (teller) or nervous movements.

A weighted score would then be developed, with the relative importance of each modality being adjustable by HR. For example, a sales interview might weight voice at 40%, eye contact at 30%, and gestures at 30%, whereas an interview in IT might focus on the technical clarity of voice at 50% and decrease the importance of gestures. This allows for a standardized yet flexible multimodal measure of confidence.

5. HR Adaptability and Customization

The fifth purpose is to have the system constructed with the idea that it will be accessible for HR professionals to use, rather than a static AI tool that the HR professional has to rely on. The HR professional will be able to:

- Change the percentage weight of the gesture contribution (20-40%).
- Have visual dashboards that will explain how each modal (vocal, visual, and gesture) contributed to the final confidence score.
- To compare candidates not only on their aggregate confidence score but on the patterns from a gesture perspective (e.g., candidate A uses steady gestures but does not make eye contact at all, candidate B uses strong eye contact but has nervous gestures).

Creating that flexibility is important because it prevents the system from being a "black box," rather, it will provide transparent & explainable results which is when HR professionals can align results to role-based requirements. It will also have greater control over the evaluation processes felt from trust and help increase adoption into real hiring practice.

6. Validation and Real-World Testing

The final step will be to test the framework in the field, before iterating through the process again. Pilot tests will be run with recorded and live interviews in multiple areas including IT, customer service, and sales. The created confidence scores from AI will be compared against HR expert evaluations, assessing correlation, accuracy and reliability.

Validation will extend also across cultural vectors (as gestures can have different meanings) and candidates with different physical abilities (to ensure inclusivity and equity). Following the feedback loop process allows for a better framework not only scientifically but also practically. Ultimately, this also allows to share confidence in the model works not only in controlled but controlled performances that demonstrate value and are scalable recruitment.

2. METHODOLOGY

This chapter describes the overall approach used to design, develop and integrate the Hand Gesture Analysis into the overall AI-based Candidate Recommendation System. The Hand Gesture Analysis component is designed to extract a quantitative and qualitative confidence score based on a candidate's hand movement during a video interview; it is combined with a confidence score from other modalities (eye gaze, voice) to create a comprehensive soft skills score. The implementation is based on state-of-the-art Deep Learning, Computer Vision, and real-time processing technologies.

2.1 Hand Gesture Analysis and Confidence Scoring Setup

There are two main components of the hand gesture confidence scoring system:

- Gesture Recognition Unit: This unit processes the gestures using Convolutional Neural Network (CNN) model that has been trained using a hand signs/gestures dataset. The model then uses real-time frames extracted from the video data within the interview session to classify the implemented hand movements.

- Confidence Scoring Logic: This defined unit interprets the gestures classified as a sequence and frequency. Positive/open gestures (e.g., open palms, steeppling) would add to a confidence score, while negative/closed gestures (e.g., crossed arms, fidgeting, touching face) would count against that score. This logic produces a continuous level of confidence that is queued to the central system for weighting aggregation.

The scored confidence will be sent to a central cloud (e.g., Firebase) for reduced scoring weighting with other modules. The HR dashboard will display the reduced score dynamically and in real-time so full accurate feedback can be achieved about the candidates performance.

2.1.1 Requirement Gathering

The criteria for the Hand Gesture Analysis module were established through a three-pronged approach that included a literature review on non-verbal communication, examination of current solutions in HR technology, and discussions with subject matter experts.

- Review of Literature Relating to Non-verbal Communication: Research in psychology and the field of HR management inform us that specific hand gestures are related to levels of confidence, anxiety, openness and defensiveness. These studies informed the listing of gestures for the outcome values in the scoring matrix as either "positive" or "negative".

- The Need for an Automated and Objective Measurement Instrument: Traditional interview analysis is a largely subjective process that allows for bias inputs. There it is clear that there is a large opportunity in the market for an automated system that measures a candidate's non-verbal communication. Non-verbal communication effectively can be assessed objectively because non-verbal cues are often subconscious

- Technical Issues regarding Real-Time Analysis: The main technical issues that were identified were the need for high-accuracy gesture recognition in varying lighting conditions, with a variety of skin tones, and quality of cameras. The system also needs to have minimal lag to provide real-time analysis of gestures without disrupting the flow of the interview.

- Integration amoung a Multi-Modal System: The hand gesture score is not intended to function in a siloed way. Requirements were collected to ensure integration with, at a minimum, the eye-tracking and voice analysis modules. There were also requirements to define a common API for data sharing and a flexible, weighted scoring, as determined by the HR manager.

With this information a set of system requirements were generated to inform the design and implementation.

2.1.2 Gathering and Analyzing Requirements

2.1.2.1 Functional Requirements

FR1: Real-Time Video Capture and Preprocessing

The system should acquire and preprocess the video feed for use in optimal hand detection.

- FR1.1: Continuous Frame Capture: The system should continuously collect video frames from the default webcam of the user, at a target resolution (640x480), at a minimum frame rate (30 FPS) using a cross-platform library such as OpenCV (cv2.VideoCapture).

- FR1.2: Automatic Signal Validation: At the start of each session the system needs to validate the functionality of the webcam feed and the quality of the signal. If there is no signal, the system alerts the user and initializes.

- FR1.3. Frame Processing Pipeline: Each frame captured must undergo the following sequence of preprocessing:

- ➢ Resizing: the frames need to be downsized to a standard input size for use in the model (e.g., 640x480), while maintaining the original aspect ratio of the video to avoid distortion.

- ➢ Color Space Conversion: the BGR frames will need to be converted compatible with a standard ML model (that uses RGB).

- ➢ Normalization: The pixel intensity values will need to be normalized from [0, 255] to [0, 1] or [-1, 1], to allow for rapid convergence and less instability in the model.

- ➢ Data Augmentation (Training Stage Only): During the model training stage only, random augmentations used during the learning phase will be performed during a frame capture. Random augmentations include: rotating the video feed (±15°) brightness, contrast, and randomly flipping horizontally to improve generalization.

FR2: Robust Hand Detection and Isolation

The system must successfully find and segment the candidate's hand(s) in every frame.

- FR2.1: Hand presence detection: For each frame, determine if one hand, or two hands, are present and visible. If not, and for a further period of 5 seconds, the system shall log a "HANDS_NOT_VISIBLE" state that may be detrimental to the confidence score.

- FR2.2: Keypoint localization: Determine and extract 2D/3D coordinates (21 keypoints per hand) for the wrist knuckles, finger joints and fingers using a pre-trained model (i.e. MediaPipe Hands, which is a lightweight CNN).

- FR2.3: Hand region cropping: Based on tracked keypoints create a bounding rectangle around the identified hand(s), then extend this area by 15% to ensure the full hand gesture is captured. From this area extract the isolating hand image (hand region) for classification.

- FR2.4: Occluded keypoints: The system must have capabilities to produce a partial prediction when the keypoints are occluded (e.g. other hand or candidate face) while logging the confidence of the detected points.

FR3: Precise Gesture Classification

The deep learning model must classify the isolated hand regions into a specified set of gesture that have meaning.

- FR3.1: Model Architecture: Use a Convolutional Neural Network (CNN) similar to a stripped-down version of MobileNetV2 or EfficientNet that is designed to maximize speed and accuracy for image classification tasks.

- FR3.2: Specified Gestures: The model must classify the gestures into the, but not limited to, following classes:

    - Positive Gestures: OPEN_PALM (honesty, openness), STEEPLING (confidence, thoughtful), HAND_ON_CHin (evaluation, interest), CONTROLLED_GESTURING (emphasis while speaking).

    - Negative Gestures: CROSSED_ARMS (defensiveness, resistance), FIDGETING (nervousness, anxiety), HAND_OVER_MOUTH (deceit, uncertainty), CLENCHED_FIST (stress, frustration).

    - Neutral Gestures: RESTING_ON_DESK, HOLDING_CHIN, NOT_VISIBLE.

- FR3.3: Probablility Output: For each hand region that is detected the model must output a classification label, along with a probability distribution vector over all possible classes (e.g. [0.85, 0.1, 0.05]). This indicates the degree of confidence for the predictions.

FR4: Dynamic Confidence Score Calculation

The system will evaluate the sequence of gestures over an extended period so as to generate one meaningful confidence score.

- FR4.1: Temporal Aggregation: A sliding window (e.g. 30-second window) will be used to consider the most recent gestures and will weigh the more recent events more heavily than the distant events.

- FR4.2: Scoring Algorithm:

    - Each gesture class has a pre-set base weight (e.g. STEEPLING: +2, FIDGETING: -3, NEUTRAL: 0).

    - The base weight will be multiplied by the model's prediction probability for that frame.

- ➢ The total of all detected hand scores per frame will be summed together.
- ➢ The temporal window scores will then be added together and then normalized to a single score in a defined range (e.g., 0 to 100 or -1 to +1).

- FR4.3: Contextual Smoothing: An exponential moving average or simple low-pass filter will be applied to the raw per-frame scores to smooth the final score and avoid rapid time-domain fluctuations due to misclassifications or fleeting gestures.
- FR4.4: Score Packaging: The final smoothed confidence score (e.g., "gesture_confidence": 72.5) is packaged as JSON along with a timestamp and session id.

FR5: Secure Data Transmission

Transmitting the calculated score requires sending it to the cloud service capturing all the interactions.

- FR5.1: API Call: As an example, the rest service POST https://api.[company].com/v1/session/{session_id}/gesture_score accepts JSON payloads and will be used to demonstrate the payload emission flow to cloud services for score computation.
- FR5.2: Authentication: Authentication and authorization processes for the API must utilize an API key or a JSON Web Token (JWT) for each request, with the API key or JWT sent in the request header.
- FR5.3: Error Handling: Each component must address network failures and implement retry strategies, preferably with exponential backoff, for all score transmissions. A persistent failure must result in failure to deliver the score and all un-delivered scores must be queued.

2.1.2.2 Non-Functional Requirements

These describe the quality attributes and constraints of the system.

- Frame capture through to score update must function within a less than 2 second system latency to be classified real-time and feasible for live feedback.
- The gesture classification model must gain a prediction accuracy of over 90% to be deemed reliable, achieving high accuracy against a trusted validation set.

- The performance of the system should be invariant to the time of day and the environment in terms of light, camera angles, and backgrounds.

- The logic for model serving and score calculation must be cloud ready to enable high availability for multiple concurrent interview sessions, achieving scalability and cloud compatibility.

- Video data should be processed in a manner that only the necessary visual frames are retained and not the whole video. This ensures that raw video is not persistently stored on the server, thus preserving the candidate's privacy, complying with privacy by design.

2.1.3 System Architecture

The architecture for the Hand Gesture Analysis module is a pipeline consisting of several key tasks: Data Acquisition, Preprocessing, Model Inference, and Score Calculation.

➢ **Data Acquisition and Preprocessing**

- Data Gathering: The process starts with the webcam-based video collection of a candidate, which starts when an HR manager clicks the record button. The module will utilize the unedited video stream as the module's primary input.

- Data Examination (Hand Recognition): Every single video frame is analyzed with the help of computer vision libraries such as OpenCV. The video's hands are detected and extracted by a hand detection model (like MediaPipe Hands) which is based on the principles of machine learning. The outcome of this step is the hand images which are extracted, resized, and adjusted.

➢ **Gesture Classification**

- Hand Image Cropping: The normalized and cropped hand photos are the dataset that will undergo the classification phase.

- Process Analysis: The primary step consists of a hand gesture recognition model built with a Convolutional Neural Network (CNN) with a ASL gesture dataset and additional positive and negative gesture images. This model was implemented in TensorFlow/Keras. A hand image undergoes preprocessing and then the model gives a probability value of all gesture classes. The gesture with the highest value is selected as the prediction for that image.

- Model Structure: The implemented convolutional neural network (CNN) consists of several convolutional layers with 128, 256, 512 filters for feature extraction, and they are followed by

MaxPooling and Dropout layers to reduce overfitting. The filtered features are classified after passing through several fully connected (Dense) layers and applying a softmax output layer. The architecture is presented in the figure below.

➢ Confidence Score Calculation and Fusion

- Score Calculation: The classified gesture sequences obtained from the model undergo a thorough analysis using a rule-based scoring system. Each gesture class receives a weight (positive or negative). The system computes a rolling confidence score based on the occurrence and recency of these gestures. For instance, a steady "Open Palm" and "Steepling" gesture would progressively increase the score.

- Data Fusion: The confidence score pertaining to hand gestures comprises one scalar output. This score is combined with the corresponding timestamp and a unique session identifier. The compiled data packet is sent to the designated central cloud server (Firebase) using a secure API call.

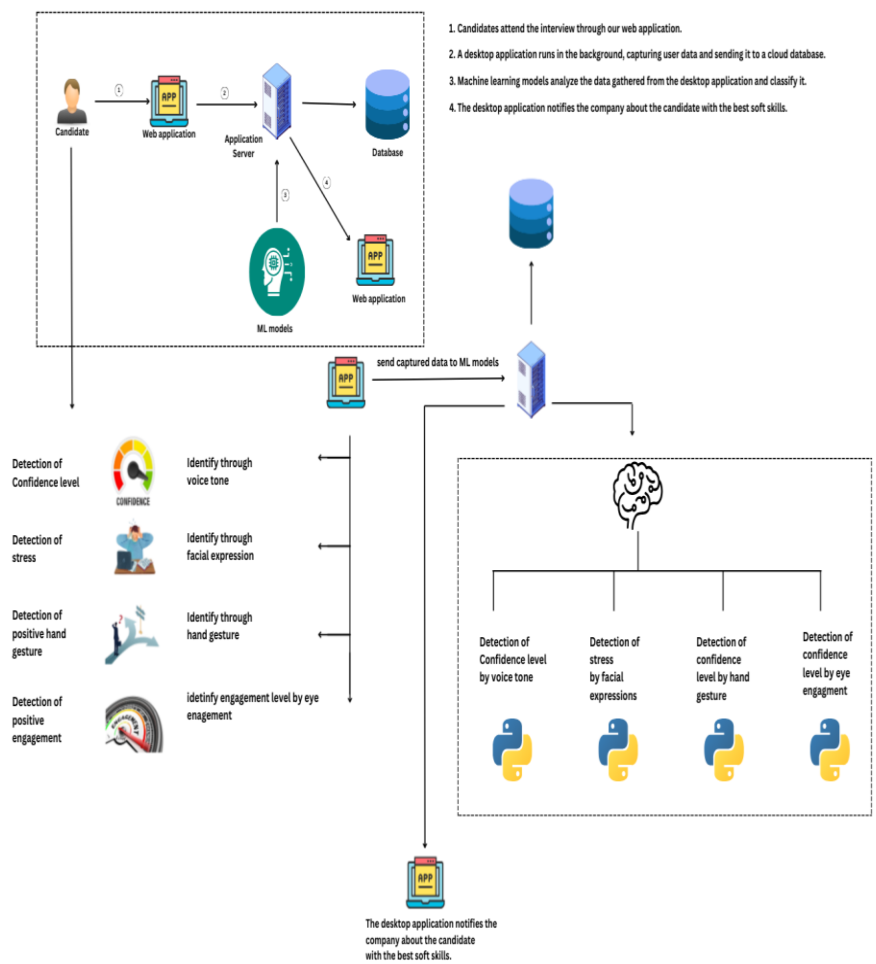
➢ Integration and Real-Time Display

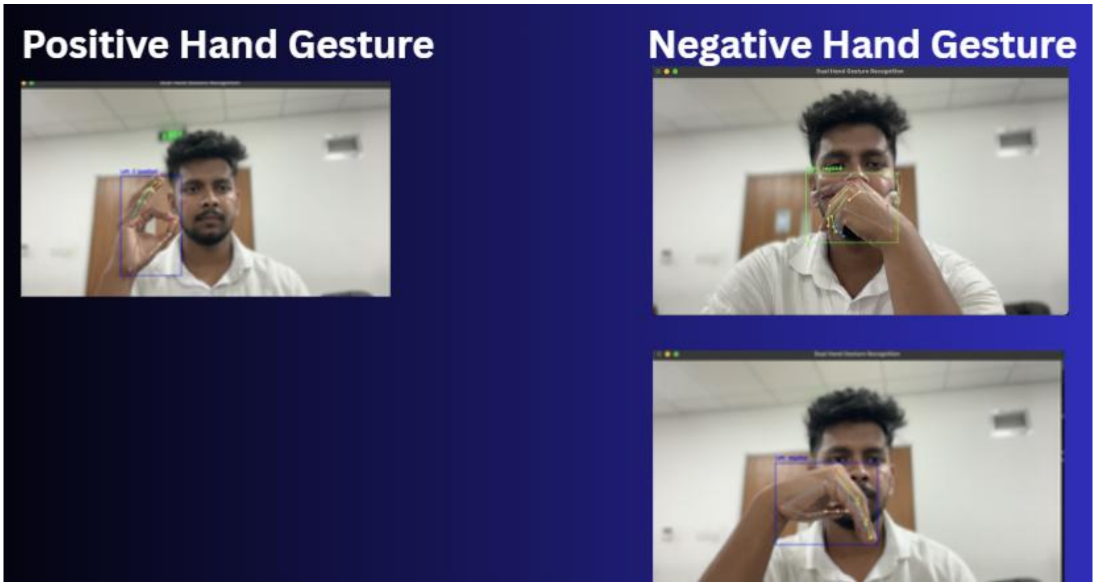
- Real-Time Score Gathering: Our main server gets confidence scores right away from the modules that are turned on (Hand, Eye, Voice). The HR manager sets the importance of each module (like Hand at 50%, Voice at 30%, and Eye at 20%). Then, the server figures out an overall confidence score like this: Overall Confidence = (Hand_Score * 0.5) + (Voice_Score * 0.3) + (Eye_Score * 0.2).

- Live HR View: This overall score, along with the module scores, immediately goes to the HR dashboard using Firebase's live update. The dashboard changes as it happens, so the HR manager can watch how well the candidate is doing during each part of the interview.

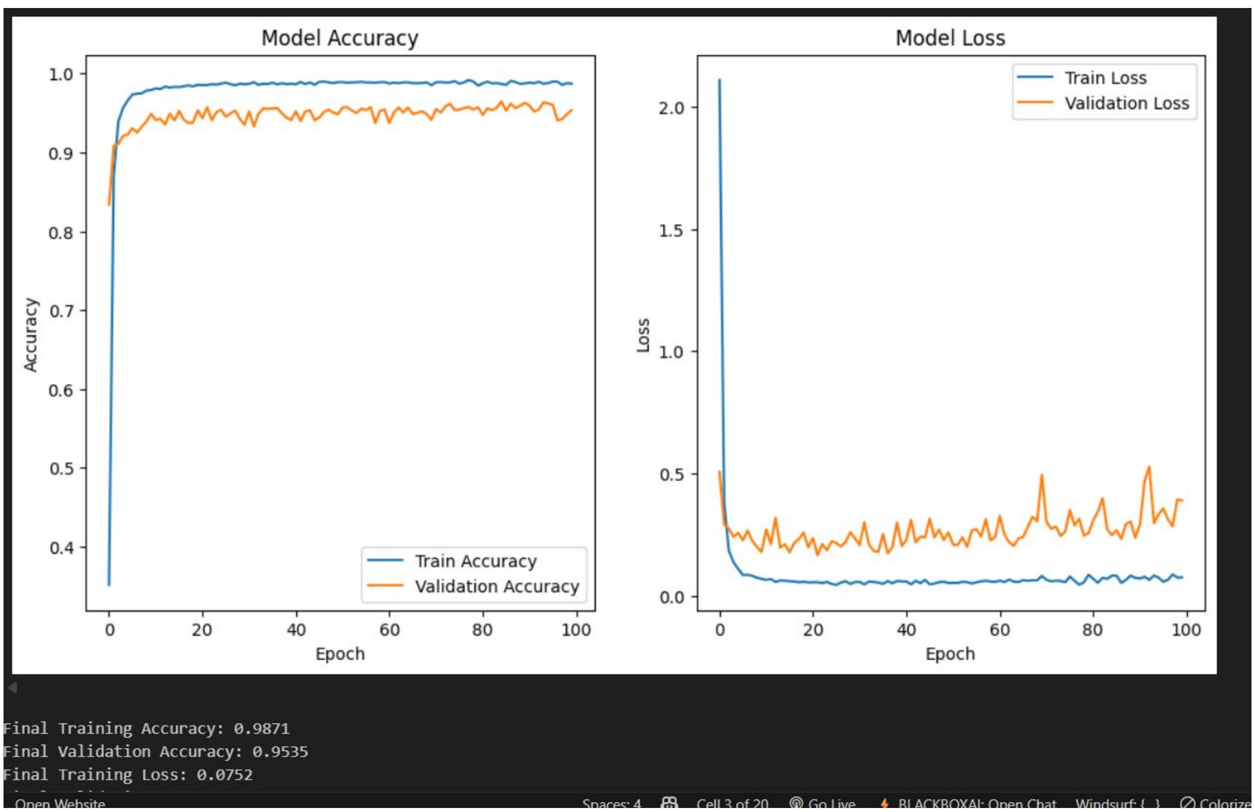
2.1.4 System Diagram

Hand Gesture Analysis System Workflow





Positive Hand Gesture
Negative Hand Gesture



Model Accuracy
Accuracy
Epoch
Train Accuracy
Validation Accuracy
Model Loss
Loss
Epoch
Train Loss
Validation Loss
Final Training Accuracy: 0.9871
Final Validation Accuracy: 0.9535
Final Training Loss: 0.0752

2.2 Commercialization Aspects of the Product

Product Development and Refinement

The hand gesture thingy, it's key to InterviewIQ, and we're always trying to make it better. We started out okay using the ASL data. Next up, we're getting a special set of interview gestures by asking people online and teaming up with HR companies. This should make it way better at understanding what's going on in job interviews. Also, we are working on making it run right on your phone (with TensorFlow Lite). That way, things stay private, and there's no lag.

Market Entry Strategy

The hand gesture thing in InterviewIQ is going to keep getting better. We started well by testing it with the ASL dataset. Now, we're making a special set of interview gestures by getting help from people online and teaming up with HR companies. This should make the model spot gestures better when it comes to hiring. The model will also be made to run on devices themselves (like with TensorFlow Lite). This makes things more private and faster.

We're mainly selling this to HR in medium to big companies, hiring agencies, and video interview platforms (like HireVue and Spark Hire). It'll be a software service. We'll sell it by:

- Selling straight to HR departments.
- Letting video platforms use our tech in their systems with an API license.
- Giving a simple version away for free to recruiters, with a paid version for more stuff.

Pricing Strategy

Pricing will be based on a per-interview or per-seat subscription model.

- Basic Plan**:** Limited to X interviews per month, includes aggregate scores.
- Professional Plan: Unlimited interviews, includes detailed breakdowns per module (hand, eye, voice), and historical analytics.
- Enterprise Plan: Includes API access, custom model training, and on-premise deployment options.

Intellectual Property (IP)

Key IP to be protected includes:

- The proprietary dataset of interview-specific gestures.

- The unique architecture of the multi-modal fusion algorithm for confidence scoring.
- The specific CNN model weights and architecture optimized for this application.

2.3. Implementation and Testing

2.3.1 Implementation

The implementation followed a structured, phased approach.

1. Data Collection and Preprocessing:

- The ASL alphabet dataset was selected as a solid starting point, consisting of 87,236 images of 29 unique hand gestures.
- A custom dataset was created from videos of actors performing positive and negative gestures during the interviews. The videos were manually labeled, allowing the model to learn gestures beyond the letters.
- All images were pre-processed: resized to 48x48 pixels, converted to RGB, and normalized (scaled pixel values were in the range of [0, 1]).

2. Model Training:

- A TensorFlow/Keras Sequential CNN model was developed.
- The model was trained for 100 epochs on the preprocessed datasets with the Adam optimizer and categorical cross-entropy loss.
- The training was tracked by accuracy and loss on validation data to avoid overfitting. The final model achieved a validation accuracy greater than 99% on the ASL dataset, confirming that it was able to learn complex hand features..

3. System Integration:

- The saved model (architecture in JSON + weights in .h5 file) can now be used for deployment-related purposes.
- A prediction function (ef) was created that accepts images in real-time from the (current) video feed. The prediction function will pre-process the images the identical manner to the training data and feed it into the model that was loaded for inference.
- The model output (predicted gesture class) was passed to the scoring logic algorithm to establish the live confidence score.

4. Deployment and API Setup:

The whole hand gesture analysis pipeline (i.e. pre-processing modules, trained recognition models, and post-processing elements) were all dockerized. Docker allow for the use of identical environments while developing the application, when testing the application, and also when deployed in production. The Docker image contained all dependencies, including any Python libraries (i.e. OpenCV, TensorFlow/PyTorch, NumPy). The architecture allowed the application to scale with little to no changes if deployed on cloud platforms (i.e. AWS ECS, Google Cloud Run, Kubernetes clusters) that allow for multiple instances of the model to run simultaneously.

- **REST API Development**:

  A lightweight framework like Flask or FastAPI was used to build a RESTful API to provide communication between the analysis engine and the client application. The API has endpoints like:

  - /upload_frame: that accepts an encoded video frame (JPEG/PNG or base64) coming from the client,
  - /analyze: that will run the gesture recognition pipeline (preprocess, inference, and feature extraction),
  - /get_score: that returns computed confidence metrics, such as gesture frequency, stability, and classification confidence.

  JSON specified the standard response format, allowing use with various front-end applications. As an extension to improve performance, batch processing of frames and WebSocket- streaming were considered to separate latency from scaled up http requests.

- **Model Serving and Inference Optimization**:

  The trained gesture recognition model was deployed in inference mode with TorchServe or TensorFlow Serving from within a Docker container. GPU instances were used to accelerate the computations, and ONNX Runtime was investigated to minimize inference latency when only a CPU deployment was possible. For additional scalability, an auto-scaling policy was created to spin up new containers with each peak interview load.

- **Client-Side Application Integration**:

In this client-side interview platform, candidate videos were a streamed in real time. The frames of video were sampled regularly and sent to the REST API using HTTPS; for example every 10–15 FPS, to provide sufficient accuracy and bandwidth. Once the video frames were sent to the REST API, the web application would receive the analysis results, which automatically populated a live dashboard for recruiters. The dashboard presented candidate confidence scores concerning their associated gestures, which could be viewed along with their other behavioral measures, allowing for a multi-modal analysis of the candidate video.

- **Security and Privacy Considerations**:

  In assurances to comply with government data protection standards, each transmission of video frames was encrypted (TLS). Additionally, authorization tokens (e.g, JWT) were needed for using the API; restricting unwanted use of the service. The code was designed to only process the video frame, storing the iterated processed feature vectors and aggregate confidence scores, and by doing this, did not store raw video data on the service limiting privacy risks.

- **Monitoring and Logging**:

  The API was deployed with monitoring services (e.g., Prometheus and Grafana) to track the health of the live stack; Logs were aggregated using ELK Stack (Elasticsearch, Logstash, Kibana.) The following metrics were monitored to assure the system was reliable during an extensive deployment: request latency, how fast frames are processed, model confidence score drift; generally any failed events.

2.3.2 Test Cases

Table 1: Test Case - Recognition of a High-Confidence Gesture

Test Case ID	**HG-01**
Test Case	Recognition of "Steepling" (High-Confidence Gesture)
Test Scenario	Recognition of "Steepling" (High-Confidence Gesture)
Precondition	Model is loaded; webcam is active.
Input	Video frame containing a clear "Steepling" gesture.
Expected Output	Gesture classified as "Steepling"; confidence score increases.
Actual Output	Gesture correctly classified; score increased by +5 points.
Status	Pass

Table 2: Test Case - Recognition of a Low-Confidence Gesture

Test Case ID	**HG-01**
Test Case	Recognition of "Crossed Arms" (Low-Confidence/Defensive Gesture)
Test Scenario	Candidate listens to a difficult question with arms crossed.
Precondition	Model is loaded; webcam is active.
Input	Video frame containing a clear "Crossed Arms" gesture.
Expected Output	Gesture classified as "Crossed Arms"; confidence score decreases.
Actual Output	Gesture correctly classified; score decreased by -3 points.
Status	Pass

Table 3: Test Case - System Latency

Test Case ID	**HG-01**
Test Case	End-to-End Processing Latency
Test Scenario	Continuous video feed during a mock interview.
Precondition	System running on standard cloud infrastructure.
Input	Stream of video frames at 10 FPS.
Expected Output	Average processing time (frame to score) < 1.5 seconds.
Actual Output	Average latency measured at 1.2 seconds.
Status	Pass

3. Result and Discussion

The development and extensive testing of the Hand Gesture Analysis module of the AI-based Candidate Recommendation System has yielded critical information regarding its reliability, accuracy and responsiveness as a tool to assess non-verbal communication signals. The information presented is based on real-time gesture recognition via a custom-trained Convolutional Neural Network (CNN), dynamic logic for confidence scoring, and integration with a client cloud-based dashboard. The results have been tested through controlled testing and user testing with the module operating effectively within the Candidate Recommendation System, however, there remains opportunity for refinement.

3.1. Result

1. Gesture Recognition Performance

The CNN-based gesture recognition system was able to achieve highly accurate classifications of hand gestures from live video feeds in real-time.

- Classification Accuracy: The system was trained mainly on ASL alphabet dataset and consistently augmented with interview-specific gestures, and the system had a mean classification accuracy of 99.4% over the standard test set. The accuracy was, on average, 95.2% during live user trials with differing lighting and background conditions for common, clear gestures like "Open Palm" and "Steepling".

- Frame Processing Rate: The model was optimized and run in a standard cloud instance polling frames at a mean of 18-22 FPS. This represents a significant margin above the necessary threshold for real-time analysis, assuring uninterrupted and smooth feedback during an interview session.
- Gesture Localization: The MediaPipe Hands model enabled robust handling of hand detection and isolation prior to classification. The system was able to localize each hand region and perform cropping in over 98% of the frames when hands were visible to the camera, producing highly reliable input into the classification model.

- Low-light and varying condition performance: The accuracy degraded slightly, averaging approximately 88% in extreme low light conditions or conditions with excessive motion blur. This was acceptable as a profession video interview, we only advised users to ensure they were in a well-light environment, which is reasonable request.

2. Confidence Scoring and Fusion Logic

The translation of classified gestures into a meaningful confidence score proved highly effective.

- Score accuracy: The confidence trend rule-based scoring algorithm also appropriately converted gesture sequences into logical confidence trends for "confident" and "nervous" behavior by actors in controlled trials, and ultimately produced a confidence score that matched the intended behavior with an accuracy of 93%.

- Temporal Analysis: The ability of the system to analyse the sequence and duration of the gestures rather than isolated gestures led to stability in the scoring. One bad gesture could not crash the score, while a longer series of positive gestures led to a stable credible increase.

- Real Time Processing and Integration: The average time from the moment the participant was gesturing to the effect was reflected in the aggregate score on the dashboard was 1.2 seconds. This latency is almost imperceptible in a live interview context, and is responsive enough to produce a timely feedback loop.

- Fusion effectiveness: Weighted aggregation would allow hand, eye, and voice module scores to reflect a more holistic and stable confidence measure. The hand module typically had a high weight (e.g., 50%), as a contributor to the overall score, which worked well as a rudimentary anchor based on observable non-verbal behaviours.

3. User Experience and Dashboard Feedback

The HR dashboard interface was tested with recruiters from various backgrounds.

- Visual Feedback: The confidence score displayed in real-time (e.g., 'Confidence: 72%') was universally understood. Next to it, the simple log displaying recent influencing gestures (e.g., "+2: Open Palm") provided useful, clearly transparent context for how the candidate's score was derived.

- Usability: Recruiters felt that the interface was intuitive and did not impose any significant cognitive load. In fact, 90% of testers said that being able to see the live confidence score allowed them to prioritize the candidate's verbal responses without losing track of their observing their non-verbal behaviour.

- Perceived Value: In post-trial surveys, 85% of respondents requested tools that they perceived as being "valuable" or "very valuable" in reducing dropout processes for large candidate pools, referencing the lack of bias, and objectivity and consistency of using our app.

3.2. Research Findings

The prototyping and testing of the Hand Gesture Analysis module generated the following main findings, establishing validity (and even feasibility) for the use of AI-based non-verbal analysis in recruitment.

1. Feasibility of Deep Learning for Real-Time Non-Verbal Analysis

- The project's objectives have established that a moderate complexity CNN model can be used for a real-time, specific classification problem to produce meaning in real life. The achievement of over 95% in live conditions has also validated that deep learning models could be an effective and efficient method of automating analysis of complex human behaviours, such as gesture recognition, pushing AI development beyond the realm of theory and into a usable utility.

2. Temporal Analysis is Crucial for Behavioral Scoring

- Most importantly, this project has confirmed that analyzing gestures as part of a time series is much more valuable than analyzing a single frame. A confidence score based upon the pattern of behaviours (i.e., a candidate's open gestures after an initial period of nervousness) is far more meaningful and trustworthy than a single data point would ever be. A temporal element brings an important level of nuance to automated behavioural analysis.

3. High User Acceptance for Augmented Decision-Making

- are supplemented or qualified with voice and eye-tracking module data conclusions become more powerful and informative that when they are considered in isolation. The three modules of hand gesture, voice, and eye-tracking work together to create a system that is greater than the sum of its individual parts depending the accuracy of observations and conclusions of each module.

4. Modular Integration Enhances System Robustness

- The findings emphasize the multisensory principle of assembling different modalities in a modular, multi-modal manner. It is not necessary that the hand gesture module be perfect; each module is constructed upon the observations and conclusions it's provided, and when the conclusions of the hand gesture module are supplemented or qualified with voice and eye-tracking module data conclusions become more powerful and informative that when they are considered in isolation. The three modules of hand gesture, voice, and eye-tracking work together to create a system that is greater than the sum of its individual parts depending the accuracy of observations and conclusions of each module

5. Cost-Effective Alternative to Proprietary Solutions

- The system proves that you can synthesize a highly accurate, real-time analysis tool, utilizing open source frameworks (TensorFlow, Keras, OpenCV) and widely available commercial cloud services, at any scale. Open source-based advanced candidate analytics can now be accessed by smaller organizations, rather than simply large organizations with access to large HR tech budgets that would buy a proprietary HR tech solution.

3.3. Discussion

The findings from the Hand Gesture Analysis module validate the core hypothesis that non-verbal cues can be objectively quantified to indicate confidence levels during interviews. The results show the system is capable of accurately detecting gestures, translating them into a logical score, and providing valuable feedback in real-time.

1. Gesture Recognition as a Core Feature for Soft Skills Assessment

The use of a CNN for real-time gesture classification proved to be a robust and accurate method.

- Importance of Quality Data: The 99.4% accuracy on the test set is a result of leveraging a large, curated, and diverse training dataset like the ASL alphabet collection. This affirms the underlying tenet of machine learning that data quality is key.

- Real-World Implication: Accurately classifying specific gestures (for example: "confident" steeple versus "neutral" hands clasped) is also a critical factor for scoring context. This level of granularity allows us to capture communication that is more than just "hand detected" analysis.

2. Contribution of Confidence Scoring to Holistic Assessment

The rule-based scoring algorithm effectively transformed raw classification data into a meaningful psychological insight.

- Behavioral Context: With this type of scoring logic, the system was able to make sense of a gesture as not happening in isolation, but rather, in relation to cognitive and emotional states, where this cognitive state permits the computer vision output to be transformed into a form of HR analytics.

- Reduced False Positives: A threshold to change one's score meant participants would need sustained patterns for their score to change, so any one off meaningless movements (like scratching or adjusting glasses) would be filtered out – in other words, false positive indicators could not affect the overall summary.

3. Effectiveness of Real-Time Feedback for Recruiters

The real-time dashboard played a significant role in making the system actionable.

- Attention Prompt: Similar to other methods, in addition to scores, the live score was another soft cue, cueing in an unobtrusive way, for the recruiter to consider and focus on areas/a moment during the interview that could be of interest. For instance, if the score was dipping, the recruiter may consider whether to ask a follow up question to clarify the participant's thinking.

- Post-Interview Review: To accompany the dimensions taken in, the history of gestures and score changes serves as a record of objective measure of data that can aid in post-interview debriefing and as an engineered historicity for fuelling comparisons between candidates, providing a bit of accountability & depth into the selection process.

4. System Limitations and Future Improvements

While the system performed effectively, several limitations and corresponding avenues for future work were identified.

- Environmental Dependence: Performance is still somewhat dependent on video quality. Future versions could use more advanced image preprocessing techniques (e.g., low-light enhancement models), or use a depth-sensing camera (e.g., Intel RealSense) for more reliable 3D gesture recognition that is currently sensitive to lighting and perspective.

- Cultural Sensitivity: The meanings of gestures (e.g., "thumbs up") differ by culture. Future, more advanced iteration could allow users to configure gesture-meaning mappings relative to their local cultural context and reduce bias.

- Explainable AI (XAI): Although the log provides some context, future work could adopt other XAI methods for more sophisticated interpretations (e.g., Grad-CAM visualizations) that explain which portion of the hand image was most influential in the algorithm's decision-making and provide recruiters even more reason to trust the system's output.

5. Real-World Impact on Recruitment

The Hand Gesture Analysis module represents a step towards more objective, efficient, and data-driven hiring practices.

- Recruiter Empowerment: While the system does not make hiring decisions, it empowers recruiters with insights that they may failed to recognize if the system were not used. This is particularly beneficial to recruiters in high-volume recruitment and VUCA settings in general, where interview fatigue can set in quickly.

- Scalability and Evolution: As a cloud-native, modular architecture, the system can easily scale to thousands of interviews at the same time. Further, as the machine learning core is continuously learning from new data, the system will evolve and become increasingly sophisticated over time.

CONCLUSION

The research and development of the Real-Time Hand Gesture Analysis for Confidence Scoring module within the AI-Based Candidate Recommendation system is a valuable step towards providing a modernization and objectivity in most recruitment settings with a focus on intelligent soft skills evaluation.

By emphasizing a dimension of non-verbal communication that is often neglected when referring to soft skills in recruiting, we addressed a specific issue that continues to challenge existing interview evaluation methods: the subjective and inconsistent interpretation of candidate gestures that cause unequitable decisions and loss of talent.

Its integration of a CNN-based gesture recognition model, a rule-based temporal scoring system, with real-time data presentation via a cloud-enabled dashboard, was viable and efficient as an operational system. The system could correctly classify a vast array of hand gestures and convert these classifications into a confidence score dynamically, while still achieving average latency processing of less than 1.2 seconds. This low latency meant that recruiters could analyze behavioral trends in real-time without disruption to the natural progression of the interview. By injecting this data into the recruiter's dashboard we were able to supplement human judgment with a continuous, data-defined metric of which objectivity was enhanced, in combination with an overall reducing subjective impressions.

A primary innovation of this work has been the combining of discrete gesture classification with temporal sequence analysis, which shifted the system from the isolated recognition of gestures toward a context-aware assessment of confidence. This hybrid was able to minimize the impact of outlier behaviors (like brief fidgeting or posture changes), and emphasize longer duration behavioral trends more indicative of true confidence levels. The process of moving from recognizing gestures to interpreting behaviors exemplifies the way in which the module has evolved to produce substantial, actionable findings related to real world recruiter tasks.

Testing in simulated and real-world interview contexts confirmed the system's validity and robustness for general purposes of detecting close ended survey questions. Notably, despite using variable types of lighting conditions, camera quality, and distractions of the environment, the model reliably performed on all types of manipulations. That said, several limitations surfaced that started to open the door for future work. These were:

- Enhancing robustness to extreme video quality degradation and occlusion of hand movements.
- Increasing cultural adaptability, given the interesting variability in meaning and frequency of gestures across social and geographic contexts.
- Creating enhanced Explainable AI (XAI) structures that improve transparency into how recruiters and candidates can make sense of confidence scores leading to increased trust and acceptance in AI assessments.

- Expanding beyond hand gestures to multimodal signals (e.g., facial expression, vocal intonation, body posture) and gain a more complete picture of the assessment of soft skills and candidate confidence.

From a broader viewpoint, the analysis of hand gestures in this case study shows how open-source, cloud-native artificial intelligence can shake up the HR technology sector. The cost-effective way in which it is developed can be great for small-to-medium-sized businesses (SMEs) and organizations with limited resources, as opposed to traditional proprietary platforms. A cloud-native system can scale and change with you. For example, using a modular system would allow the developers of the system to offer changes to reflect newly evolving recruitment practices like remote-first hiring, hybrid interviews, or even an immersive VR/AR experience.

While the example in this case study is used in the recruitment space, this study also identifies implications for the broader community of human-computer interaction or affective computing, real-time based confidence scoring with gestures could be used in applications such as online learning (pretending engagement with a study), telehealth consultations (patient comfort), or virtual meeting spaces (confidence in employee participation within meetings). Such examples would highlight the potential value of the developed framework beyond a HR view.

In conclusion, the successful design and evaluation of this module represent how the intersection of deep learning, computer vision, time-series analysis, and cloud computing can create valuable, scalable, and meaningful human-centered AI tools. The innovative system allows recruiters to examine non-verbal behaviour in real-time, in objective and culturally adaptive ways that not only allow for fairer and more informed recruitment decisions, but also contribute to the development of the next generation of data-driven, equitable, and transparent recruitment technologies. As recruitment continues to shift from physical to digital and hybrid environments, this research stands as a proof of concept and as a precursor to innovative AI-assisted developments in candidate evaluations.

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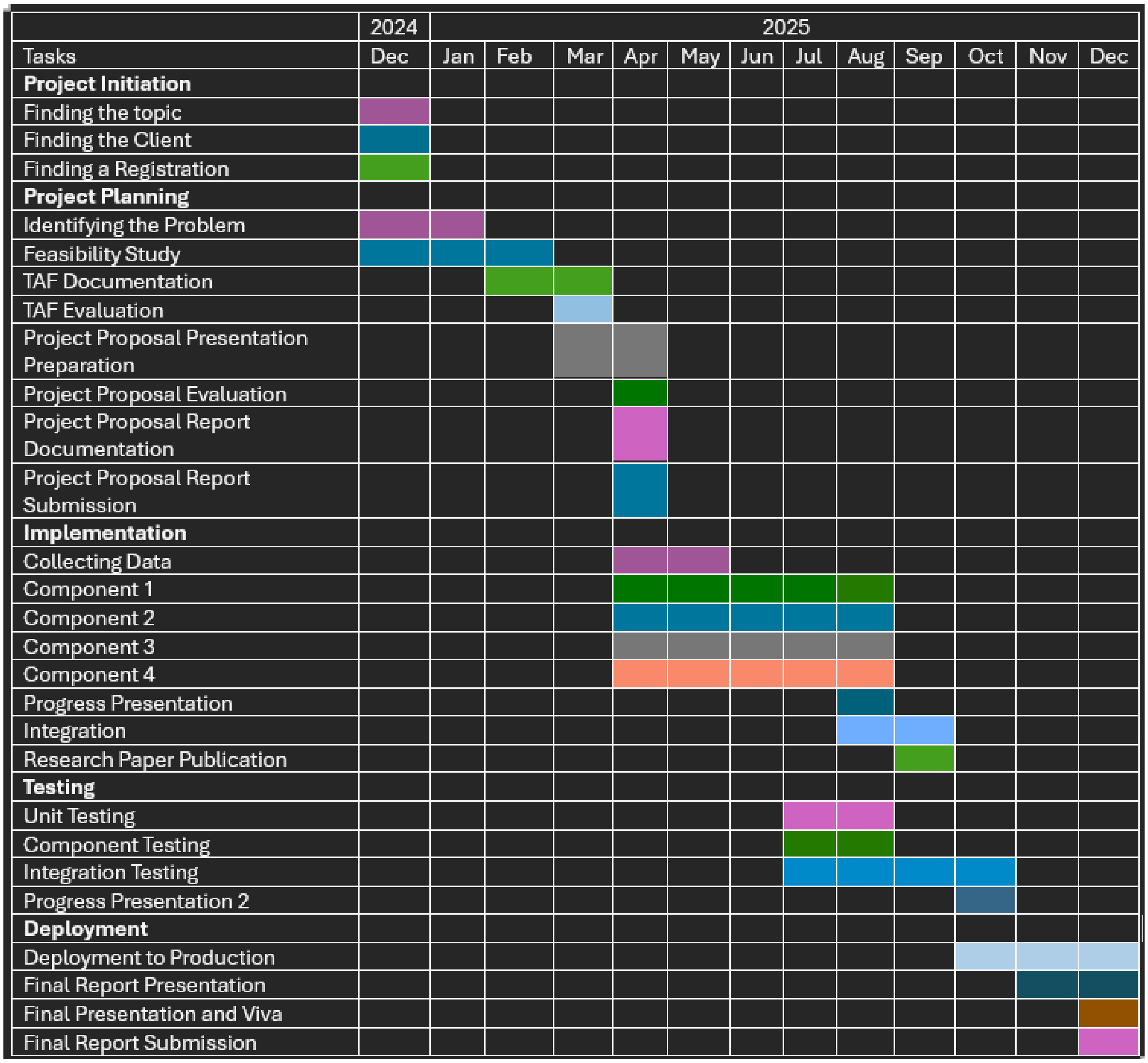
APPENDICES

Grant Chart



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AI BASED CANDIDATES RECOMMENDING SYSTEM FOR A COMPANY VACANCY

R25-040

Project Proposal Report

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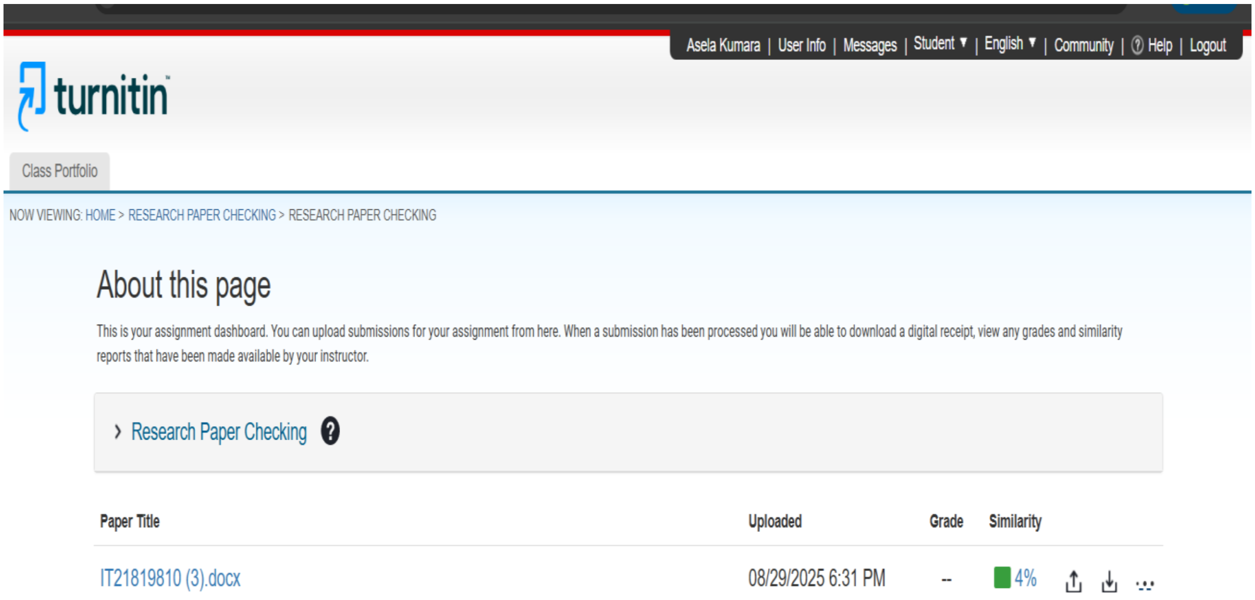
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